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(See 6.5)

MILITARY SPECIFICATION

ALIDADE, TELESCOPIC, MARINE: MARK 8 AND ADAPTER RINGS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the marine telescopic alidade Mark 8, Mod-0 used on magnetic compasses and gyrocompass indicators having nominal 6 or 7-1/2 inch diameter cards. Adapter rings for use with the alidade and an alidade stowage case are also covered.

1.2 Classification.

1.2.1 Adapter rings for the alidade shall be of the following types as specified (see 6.2.1):

Type A - For 7-1/2 inch magnetic compass.
Type B - For 7-1/2 inch gyrocompass indicators.
Type C - To adapt to the type A and B adapter rings.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116 - Preservation-packaging, Methods of.
MIL-C-675 - Coating of Glass Optical Elements (Anti-Reflection).
MIL-S-901 - Shock Tests, HI (High-Impact), Shipboard Machinery Equipment and Systems, Requirement for.
MIL-P-15024 - Plate, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-M-15071 - Manual, Technical, Equipments and Systems Content, Requirements for.
MIL-E-16400 - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for.
MIL-I-17214 - Indicator Permeability, Low-Mu (Go-No-Go).
MIL-F-21424 - Filter, Polarizing (For Optical Instruments).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).

2.1.2 Other Government drawings and publication. The following other Government drawings and publication form a part of this specification to the extent specified herein.

DRAWINGS

DEPARTMENT OF THE NAVY

NAVAL SEA SYSTEMS COMMAND

ALIDADE, TELESCOPIC MARK 8 MOD 0

5002390 - General Assembly and Optical Arrangement.
5002391 - Eyepiece Assembly and Details.
5002392 - Housing Assembly and Details.
5002393 - Prism Assembly and Details.
5002394 - Filter Assembly and Details.
5002395 - Objective Assembly and Details.
5002396 - Auxiliary Optical System Assembly and Details.
5002397 - Bearing Rings Assembly and Details.
5002398 - Case.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND

NAVSHIPS 324-0418 - Telescopic Alidade Marine 6-Inch, Mark 6, Mod 1 and 7-1/2 inch, Mark 7, Mod 0.

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(Copies of drawings and publication required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein.

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, N.W., Washington, DC 20036.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Samples for first article inspection. When specified (see 6.2.1), a sample unit shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Description. Alidades shall consist of a reticle-equipped terrestrial telescope for viewing distant objects and an auxiliary optical system for reading the gyro or gyroscope card and level bubble. Adapter rings shall fit the alidade to a 7-1/2 inch gyrocompass indicator and magnetic compass.

3.3 General requirements.

3.3.1 Alidades, adapter rings, and stowage case shall be in accordance with the following paragraphs of MIL-E-16400 in addition to the requirements specified herein. Whenever a requirement of MIL-E-16400 conflicts with a requirement of this specification, the requirement of this specification shall govern. Wherever a paragraph heading is listed, all subparagraphs under that heading shall apply, except that where subparagraphs are specifically cited, only those cited are applicable.

Materials:
Fungus inert materials
Flammable materials
Metals:
Aluminum
Ferrous alloys
Nonferrous mateials (except aluminum)
Springs (material)

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Lubricants and lubrication (see also 3.3.2.2)
Painting (see also 3.3.2.3)
Protection against corrosion
Bolts, machine screws, studs, and nuts
Parts - Mechanical:
Gaskets
Features - Mechanical:
Enclosures (see also 3.4.2)
Rounded corners and edges
Temperature and humidity (see also 3.3.3)
Salt fog (see also 3.3.4)
Miscellaneous requirements:
Interchangeability and standardization (see also 3.5)
Workmanship (see also 3.7)

3.3.2 Materials. Materials shall be in accordance with NAVSEA Drawings 5002390 through 5002398.

3.3.2.1 Non-magnetic materials. Magnetic permeability shall be 1.2 or less after fabrication (see 4.6.1).

3.3.2.2 Lubricants and sealing materials. Lubricants and sealing materials shall withstand the temperature range specified in 3.3.3 and shall not cause or contribute to filmig of the optical surfaces, or support fungus growth when subjected to the temperature test of MIL-E-16400.

3.3.2.3 Finishes. Parts shall be finished in accordance with NAVSEA Drawings 5002390 through 5002398. Mold marks shall be removed to preserve the effect of a precision instrument.

3.3.2.4 Recovered materials. Unless otherwise specified herein, all equipment, materials, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3.3 Temperature. Alidades and adapter rings shall meet the requirements for extreme temperatures and exposed locations as specified for range 2 in MIL-E-16400.

3.3.4 Humidity. Alidades and adapter rings shall meet the requirements for humidity as specified in MIL-E-16400.

3.3.5 Salt fog. Alidades and adapter rings shall show no appreciable corrosion or malfunctioning of rotatable, movable, or removable parts when subjected to the salt spray test specified in MIL-E-16400.

3.3.6 Shock. Alidades shall conform to the class I, grade A, lightweight equipment shock requirements of MIL-S-901 (see 4.6.2). Following the shock test, the equipment shall pass the collimation test (see 4.6.5).

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3.3.7 Vibration. The equipment shall conform to the type I vibration requirements of MIL-STD-167-1 and shall exhibit no signs of damage or loosening of parts during and after the vibration test (see 4.6.3). Following the vibration test, the equipment shall pass the collimation test (see 4.6.5).

3.3.8 Marking. Each alidade, adapter ring, and stowage case shall have an identification plate in accordance with type A, B, or C of MIL-P-15024 and shall conform to the normal service requirements of MIL-P-15024/5, except that reversed etched plates shall not be used. Brass plates shall not be used on the alidade or adapter rings. The following information shall be shown on each identification plate:

(a) Name of assembly:
"Marine telescopic alidade" or "Adapter ring for marine telescopic alidade".
(b) Mark and Mod number of alidade or type number of adapter ring as specified (see 6.2.1).
(c) National stock number as specified (see 6.2.1).
(d) Serial number (on alidade only) as specified (see 6.2.1).
(e) Contract number.
(f) Year of manufacture.
(g) Contractor.
(h) Naval Sea Systems Command.
(i) Space for inspector's stamp.
(j) Initials "U.S.".

3.4 Detail requirements. Alidades and adapter rings shall be in accordance with Drawings 5002390 through 5002398. The performance requirements shall be as specified hereinafter.

3.4.1 Construction and assembly. Each alidade shall consist of an enclosure, optical system, bearing ring, mounting supports, handle, and level vial. the alidade shall fit the bezel of a 6-inch gyrocompass indicator and magnetic compass without causing an imbalance of the gimbaled instrument. With the use of the appropriate adapter rings the alidade shall fit the bezel of a 7-1/2 inch gyrocompass indicator and magnetic compass without causing an imbalance of the gimbaled instrument.

3.4.2 Enclosure. The enclosure shall be as specified in MIL-E-16400, except that the degree of enclosure for the alidade shall be airtight in accordance with MIL-STD-108 (see 4.6.4). The enclosure shall be filled with a dry nitrogen or dry-nitrogen air mixture to 4 pounds per square inch (lb/in^2) gage pressure at ambient temperature (see 4.6.5). The enclosure shall be cast or otherwise fabricated such that the gas passes over the maximum number of air-glass surfaces. Aluminum casting shall be impregnated to combat porosity. The enclosure shall be gasketed and sealed to prevent entrance of moisture and loss of gas during long periods of storage and under service conditions. Provision shall be made for attachment of a handle and mounting supports which shall be replaceable without loss of the internal atmosphere. The exterior surface of the objective end of the enclosure shall be machined to permit future mounting of attachments.

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3.4.3 Optical system. The general arrangement of the optical system shall conform to Drawing 5002390.

3.4.3.1 Eyepiece. The eyepiece shall be of the internal focusing type for better sealing against dirt, dust, and moisture. The focusing range shall be from at least minus 3 diopters to plus 1 diopter; scale and index lines shall indicate the true diopter setting within 1/4 diopter. The focusing mechanism shall operate smoothly without binding or excessive motion. Each whole diopter line shall be numbered. The exit pupil shall have a diameter not less than the free aperture divided by the actual magnification of the instrument. The eye distance shall be at least 22 millimeters (mm).

3.4.3.2 Eyeguard. A silicon rubber eyeguard shall be provided to exclude extraneous light from the observer's eye. The eyeguard shall be securely attached and have a cover to protect the eyepiece and exclude moisture when not in use.

3.4.3.3 Lens cover. A lens cover shall be provided for protection of the objective lens when the alidade is not in use.

3.4.3.4 Terrestrial telescope. Magnification of the alidade main field shall be 6 diameters plus or minus 2 percent. The true field of view shall be at least 8 degrees 45 minutes. The indicated bearing of an object in the field of the terrestrial telescope shall be accurate to within 5 minutes of arc with respect to the indicator or compass card. The free aperture of the objective shall be not less than 35 mm nor more than 37 mm. The definition at the center of the field shall be excellent. The central image shall be free from defects of inaccuracies in manufacture, faulty workmanship, and the presence of excessive spherical or chromatic aberration. Other portions of the field shall be as free from coma, distortion, and parallax as feasible. The limit of resolution shall not exceed 8 seconds at the center of the field and 20 seconds of arc near the edge. The primary curvature of field shall not exceed 1.3 diopters and the secondary curvature shall not exceed 3.5 diopters. The astigmatic difference shall not exceed 2.5 diopters in any portion of the field.

3.4.3.5 Auxiliary optical system. The field of the auxiliary optical system shall show clearly the reticle and at least 25 degrees of the card of the 6-inch compass, and at least 20 degrees of the card of the 7-1/2 inch compass. The level vial bubble shall show at the edge of the field.

3.4.3.6 Filter. The filter assembly shall be a two-position type which inserts either a polarizing filter in accordance with MIL-F-21424 or a clear glass compensator into the main optical system. The control shall be a two-concentric knob device with the larger knob inserting the filter and the smaller knob adjusting the density of the filter. There shall be no slippage of the filter drive during operation. The filter shall operate smoothly without uneven motion except for position detents or stops. The fixed element of the variable density filter shall be nearer to the objective, and shall be oriented to reduce to a minimum any glare from the sea surface.

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3.4.3.7 Coating. All glass-air interfaces shall be coated with a low reflective film conforming to MIL-C-675 unless the surface is in or very near a focal plane. False images, ghosts, halos, or other internal reflections from optical or mechanical parts shall not be visible through the eyepiece under normal viewing. Coating shall be capable of withstanding the environmental tests specified in 4.6.6.

3.4.3.8 Glass elements. Surfaces shall be regular to within 5 fringes maximum, with 1 fringe irregularity acceptable. Centering of lenses shall be within 0.01 mm. The equivalent focal lengths of all lenses shall not differ from nominal values by more than 1-1/2 percent. For cemented doublets or triplets, the tolerance shall apply to the whole lens. Lenses shall be mounted in cells and collimated. Prisms shall be bonded to their mounts and provision made for replacement of prism and mount as a unit for repair purposes.

3.4.3.9 Adjustment. When adjusted for a distant object, there shall be no noticeable parallax between the reticle, the level vial bubble, the distant object, and the card division. The reticle wire shall be vertical and in the center of the field.

3.4.4 Mounting supports. The mounting of the objective end of the enclosure to the bearing ring shall be by means of detachable supports which are not integral parts of the enclosure or ring. The supports shall be constructed to hold the enclosure and bearing ring in the proper relation to allow viewing of the compass card.

3.4.5 Handle. A removable handle shall be secured to the top of the enclosure. The handle shall be of such size and shape that the alidade can be lifted by a gloved hand. The handle shall be equipped with a clip-on type open type sight to facilitate training of the alidade on the target.

3.4.6 Bearing ring. The bearing rings for the alidade shall be in accordance with Drawing 5002396. The retaining device shall be capable of withstanding the abrasive action from rotation of the alidade on the compass or indicator and during removal and installation of the bearing ring, yet shall not score or otherwise damage the bezel.

3.4.7 Level vial. A level vial shall be mounted on the enclosure in accordance with Drawing 5002392. The glass vial shall be level with the mounting surfaces of the bearing ring, and shall be adequately protected against damage under the service conditions specified herein.

3.4.8 Adapter rings. Adapter rings, when required (see 6.2.1), shall be provided in accordance with Drawing 5002397.

3.4.9 Stowage case. Each alidade shall be furnished in a stowage-carrying case equivalent to that shown on Drawing 5002398. The case body shall be of a fiberglass reinforced plastic, a butadiene-vinyl plastic or a metal such as aluminum. The case shall have metal hardware conforming to the material requirements of MIL-E-16400. The case shall contain provisions for holding one type A, one type B, and one type C adapter ring in addition to the alidade.

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3.5 Interchangeability. The optical and mechanical parts of each telescopic alidade and adapter ring shall be interchangeable with the same parts of any other alidade or adapter ring of the same mark and mod, or type. Selective assembly which results in parts which are not interchangeable will not be acceptable.

3.6 Technical data. The contractor shall prepare technical data in accordance with the data ordering documents included in the contract (see 6.2.2).

3.6.1 Technical manuals. When required (see 6.2.1), type II technical manuals in accordance with MIL-M-15071 shall be furnished, except that content of the manuals shall be similar to Publication NAVSHIPS 324-0418.

3.7 Workmanship. Workmanship shall be in accordance with MIL-E-16400.

3.7.1 Optical elements. Optical elements shall be free from strain, seed, stones, striae, and bubbles. Polished surfaces shall be free from objectionable scratches, pits, stains and other defects visible in the field of view or which would impair performance. Minor material defects in any optical element or dirt, or dust specks on surfaces will not be considered sufficient cause for rejection unless they are obvious to the unaided eye when looking directly into the eyepiece of the assembled alidade against a background of brightness 3 candle power per square inch and color equivalent of 6,500 degrees Kelvin, or unless the function of the instrument is impaired by these defects. The central image of the terrestrial telescope shall be free from defects of inaccuracies in manufacture, faulty workmanship, and the presence of excessive spherical or chromatic aberration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Submission of test reports. Reports covering the inspection and tests for First article inspection and the Quality conformance inspection shall be furnished to the contracting activity when specified (see 6.2.1).

4.1.2 Material inspection. When specified (see 6.2.2), material inspection shall consist of certification supported by verifying data that materials used in the fabrication of the alidades and adapter rings and accessories are in accordance with the requirements of this specification.

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4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in MIL-E-16400.

4.4 First article inspection. First article inspection shall be conducted on one sample of the alidade and one sample of each type adapter ring, when applicable. The inspection shall consist of the examination and tests specified in table I in the order shown.

TABLE I. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
General examination	3.3, 3.4, 3.7	MIL-E-16400
Collimation	3.4.3.8	4.6.5
Optical measurements	3.4.3.4, 3.4.3.5, 3.4.3.6, 3.4.3.8	4.6.5
Coating	3.4.3.7	4.6.6
Air tightness	3.4.2	4.6.4
Collimation	3.4.3.8	4.6.5
Non-magnetic materials	3.3.2.1	4.6.1
Temperature	3.3.3	MIL-E-16400
Humidity	3.3.4	MIL-E-16400
Salt fog	3.3.5	MIL-E-16400
Air tightness	3.4.2	4.6.4
Vibration	3.3.7	4.6.3
Collimation	3.4.3.8	4.6.5
Shock	3.3.6	4.6.2
Collimation	3.4.3.8	4.6.5
Air tightness	3.4.2	4.6.4

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.5.1.1 Inspection lot. An inspection lot shall consist of all alidades and adapter rings of the same type, produced under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table II in the order shown.

4.5.1.2.1 Sampling plan. All equipments offered for delivery shall be subjected to the Group A examination and tests.

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TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph
General examination	3.3, 3.4, 3.7	MIL-E-16400
Optical measurements	3.4.3	4.6.5
Collimation	3.4.3.8	4.6.5
Air tightness	3.4.2	4.6.4
Collimation	3.4.3.8	4.6.5
Non-magnetic materials	3.3.2.1	4.6.1

4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table III in the order shown, and the sample shall be selected from inspection lots that have passed group A inspection.

4.5.1.3.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be 1.5 percent defective.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Temperature	3.3.3	MIL-E-16400
Humidity	3.3.4	MIL-E-16400
Vibration	3.3.7	4.6.3
Air tightness	3.4.2	4.6.4
Collimation	3.4.3.8	4.6.5

4.5.1.3.2 Dispostion of sample units. Sample units which have passed all the group B inspection may be delivered on the contract, if the lot is accepted and the sample units are still within specified tolerances.

4.5.2 Packaging inspection. Sample packages and packs and the inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with section 5 and the documents specified therein.

4.6 Methods of inspection.

4.6.1 Non-magnetic materials. The relative permeability when checked with a low-mu indicator conforming to MIL-I-17214 shall not exceed the permeability level specified in 3.3.2.1.

4.6.2 Shock. Alidades (including a type A or B adapter ring and a type C adapter ring), when mounted on a gyrocompass indicator, or equivalent shall be subjected to class 1, grade A, HI shock tests for lightweight

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equipment in accordance with MIL-S-901. Release during shock of locking or detent devices shall be considered cause for rejection. Equipment which has been subjected to shock test shall not be put into stock nor installed onboard ship without specific approval of the Naval Sea Systems Command (NAVSEA).

4.6.3 Vibration.

4.6.3.1 First article test. Type I vibration tests for first article shall be in accordance with MIL-STD-167-1. Vibration tests shall be conducted with the alidade mounted on either at type A or type B adapter ring and a type C adapter ring. The assembly shall be examined upon completion of the vibration test to determine conformance to 3.3.7.

4.6.3.2 Group B inspections. For group B inspection, the alidade with adapter rings shall be vibrated for 10 minutes along each of the three principal axes at a frequency of approximately 33 hertz (Hz) at an amplitude of 0.010 inch. The assembly shall be examined upon completion of test to determie conformance to 3.3.7.

4.6.4 Air tightness. Each alidade, after final assembly and adjustment shall be filled with dry nitrogen to an internal gage pressure of 8 lbs/in^2 and sealed off. After an interval of not less than 24 hours, the internal pressure of the alidade shall be measured. Any loss in internal gage pressure exceeding 0.125 lb/in^2, after correction for changes in ambient temperature, shall be cause for rejection. Prior to the final collimation check, internal gage pressure shall be released to 4 lbs/in^2.

4.6.5 Optical measurements. An examination, measurement, or check, as applicable, shall be made on each alidade to determine compliance with this specification. The alidade shall be tested for quality of image, resolution, field of view, collimation, and the other optical factors of 3.4.3. When checking factors such as resolution, diopter setting, and curvature of field, an auxiliary telescope of 5 to 7 power shall be used. When checking resolution, a black and white box type target or National Bureau of Standards test chart illuminated to 100 foot-candles may be used. For assessing correction of aberrations, a well illuminated distant pinhole subtending 8 to 17 seconds of arc may be used. Collimation shall be checked initially and finally with an internal gage pressure of 4 lbs/in^2. The effect of release of internal pressure on collimation shall be determined on the first article models.

4.6.6 Coatings. Prior to assembly, all optical surfaces having low reflective coatings shall be subjected to the following tests:

(a) Humidity - There shall be no visible evidence of film deterioration after 24 hours exposure to air at 120°F and 98 + 2 percent relative humidity.
(b) Durability - After test (a), the sample shall be subjected to the abrasion resistance test specified in MIL-C-675.
(c) Immersion - There shall be no visible evidence of film deterioration after the sample has been immersed in a 6 ounce sodium chloride (NaCL) per gallon of water solution for 24 hours.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation-packaging. Packaging shall be level A or C as specified (see 6.2.1).

5.1.1 Level A. Each telescopic alidade and adapter rings shall be preserved and packaged in accordance with method IA-14 of MIL-P-116 and as follows:

(a) The carrying case (see 3.4.9) shall be utilized as the inner container.
(b) Each alidade and adapter ring, as applicable, shall be placed and secured in the carrying case.
(c) Cushioning shall be provided to prevent movement within the carrying case.
(d) Cover shall be closed and all hardware fastened.
(e) Each carrying case shall be cushioned on all surfaces with cells, fiberboards, pads or similar media to prevent damage to the barrier bag.
(f) The exterior container shall be a fiber box conforming to PPP-B-636, type CF, class weather resistant, variety SW, grade W5C or W6C, style RSC. Box closure shall be method 5 as specified in the appendix to PPP-B-636.

5.1.2 Level C. Each telescopic alidade and adapter ring with carrying case (see 3.4.9), processed as specified in 5.1.1(b), (c), and (d) herein, shall be preserved-packaged to afford protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity for immmediate use. The contractor's normal retail or wholesale preservation-packaging methods may be utilized when such meets the requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2.1).

5.2.1 Levels A and B. Telescopic alidades and adapter rings, preserved and packaged in accordance with 5.1 and as specified (see 6.2.1) shall be packed in fiberboard boxes conforming to PPP-B-636, type CF, class weather resistant, variety SW, grade W5C or W6C, style RSC. Box closure and reinforcement shall be accomplished in accordance with the appendix of PPP-B-636.

5.2.2 Level C. Telescopic alidades and adapter rings preserved-packaged as specified (see 6.2.1), shall be packed in containers acceptable to the common carrier which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or Regulations or other carrier rules as applicable to the mode of transportation.

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5.3 Marking. In addition to any special marking required (see 6.2.1), unit package marking or labeling shall be applied as specified in MIL-P-116. Interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The telescopic alidade and adapter rings covered by this specification is intended for naval service where it is expected to withstand continuous use for long periods under military service conditions without benefit of frequent overhaul. The telescopic alidade is primarily intended to be used with the 6-inch and 7-1/2 inch magnetic compass and the 6-inch and 7-1/2 inch gyrocompass indicator for observing distant objects and determining their bearings.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Mark or type of equipment, or both, required (see 1.1 and 1.2.1).
(c) Type and quantity of adapter rings required (see 1.2.1).
(d) Quantity of first article models required (see 3.1).
(e) Whether first article test report is required (see 4.4.1).
(f) Marking information required (see 3.3.8).
(g) Adapter rings, when required (see 3.4.8).
(h) Levels of preservation-packaging and packing required (see 5.1 and 5.2).
(i) Special marking required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirements title	Applicable DID No.	Option
3.6.1	Manual, Technical Standard	DI-M-2044	Type II of MIL-M-15071
4.1.1	Test Reports	DI-I-2072	-------

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article is required, is shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should be a preproduction sample. The first article should consist of the number of units specified (see 6.2.1). The contracting officer should iclude specific instructions in all acquisition instruments, regarding arrangements of examination, test and approval of the first article.

6.3.1 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidder offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Changes from previous issue. The symbol "*" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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